ENGINEERING GRAPHICS • DIGITAL NOTES

UNIT V

Conversion of Views

Read a drawing like a picture; draw a picture like a drawing.

SYLLABUS COVERED

- ▸ Isometric projection basics
- ▸ Isometric scale, axes & circles
- ▸ Layout of orthographic views
- ▸ Isometric view → orthographic views
- ▸ Orthographic views → isometric view

1. Isometric Projection – Basics

DEFINITION

Isometric projection is a pictorial (single-view) projection in which the three principal axes of the object are equally inclined (**120° apart**) to the plane of projection, so all three axes are equally foreshortened. “Iso” = equal, “metric” = measure.

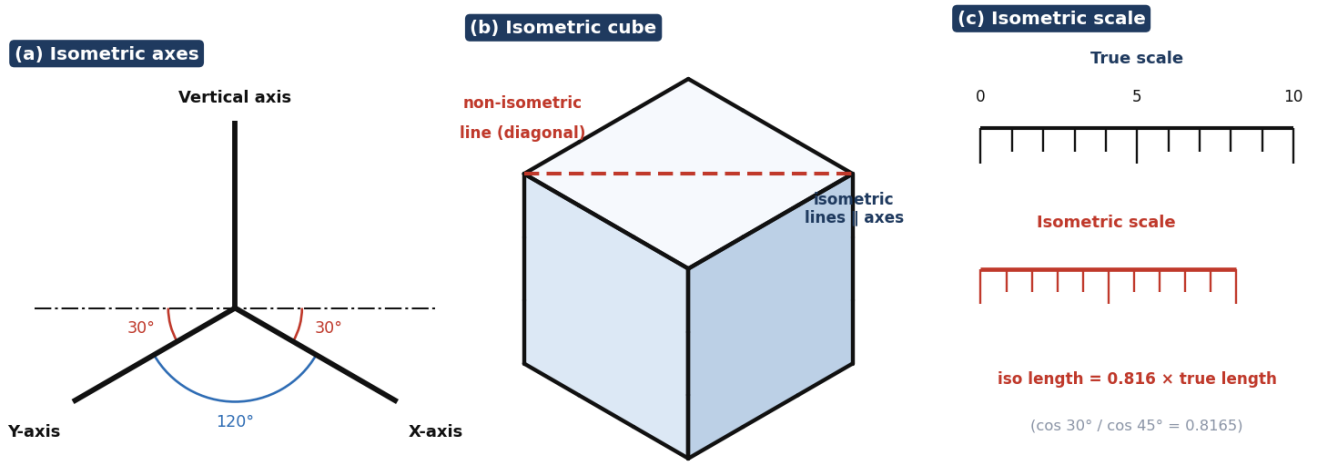


Fig. 1.1 — Isometric axes, isometric cube and isometric scale

Term	Meaning
Isometric axes	Three lines from a point at 120°: one vertical and two at **30°** to the horizontal
Isometric lines	Lines **parallel** to the isometric axes – measured in true (scaled) length
Non-isometric lines	Lines not parallel to the axes (inclined lines) – **not** measured directly; locate their end points
Isometric planes	The three faces of a cube seen in isometric: top, left (front) and right (side)

FORMULAS

Isometric scale = **cos 30° / cos 45° = 0.816** | Isometric length = **0.816 × true length** (isometric *projection*)

Isometric view / drawing uses **full true lengths** on the axes (scale 1) – the picture is 1.225 times larger but identical in shape. Normally used in practice.

EXAM TIP

Remember the three words: **Axes 120°**, **Lines ∥ axes**, **Scale 0.816**. Angles are **not** preserved in isometric – a 90° corner becomes 120° or 60°.

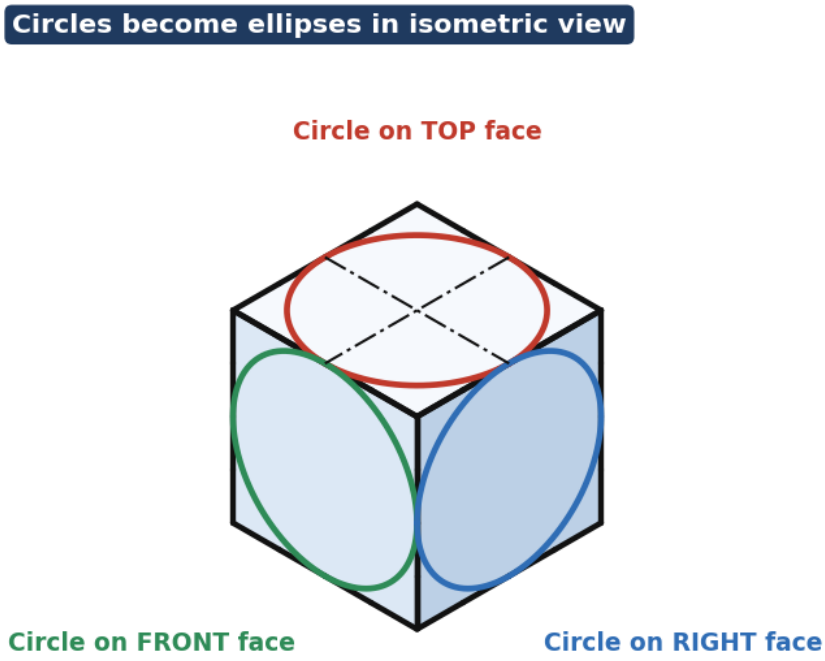


Fig. 1.2 — A circle appears as an ellipse on each isometric plane

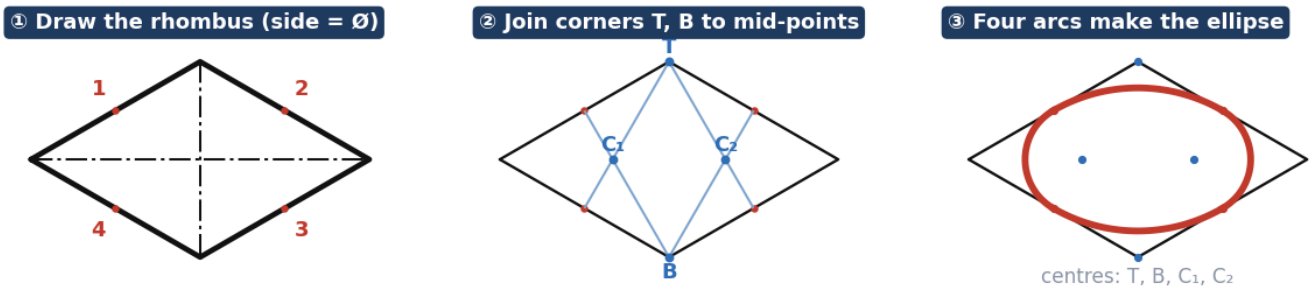


Fig. 1.3 — Four-centre method for an isometric circle

KEY POINTS

- Draw an isometric circle by the **four-centre method** inside the isometric rhombus (side = diameter): join the obtuse-angle corners to the mid-points of opposite sides – their intersections are the centres of the arcs.
- The ellipse's **major axis ⊥ to the axis normal to that plane**: on the top face it is horizontal; on left/right faces it leans along the vertical-ish direction.
- A sphere always appears as a **circle of the true diameter** (use isometric scale: diameter × 1.22 if drawn to isometric view scale).

2. Orthographic Views – Layout & Conventions

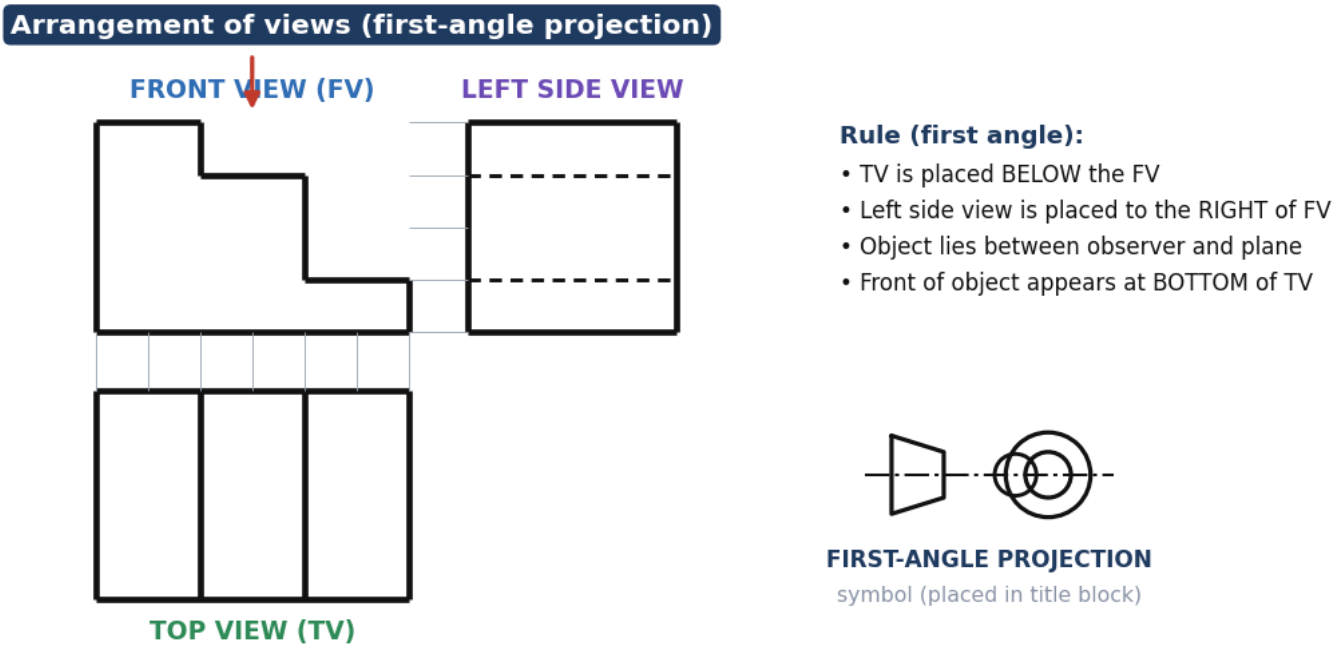


Fig. 2.1 — Arrangement of FV, TV and LSV in first-angle projection

View	Looking from	Placed	Shows
Front view (FV)	Front	Top-left of sheet	Height & width
Top view (TV)	Above	**Below** FV, aligned vertically	Width & depth
Left side view (LSV)	Left side	**Right** of FV, aligned horizontally	Height & depth

Line in a view	What it represents
Thick continuous	An edge or boundary of a **visible** face
Dashed (hidden)	An edge **behind** a face – drawn where the edge is hidden from the viewer
Chain thin	Axis of a hole / cylinder, or symmetry line
A closed rectangle inside a view	A pocket, hole or slot (seen from the side where it opens)
Single line between two faces	A **change of level / surface** (step, slope, or edge between two faces)

EXAM TIP

Projection rules: **(1)** FV & TV have the same **width**, aligned vertically. **(2)** FV & LSV have the same **height**, aligned horizontally. **(3)** TV & LSV have the same **depth** – the front of the object is at the **bottom of TV** and the **right of LSV**.

3. Isometric View → Orthographic Views

Isometric → Orthographic

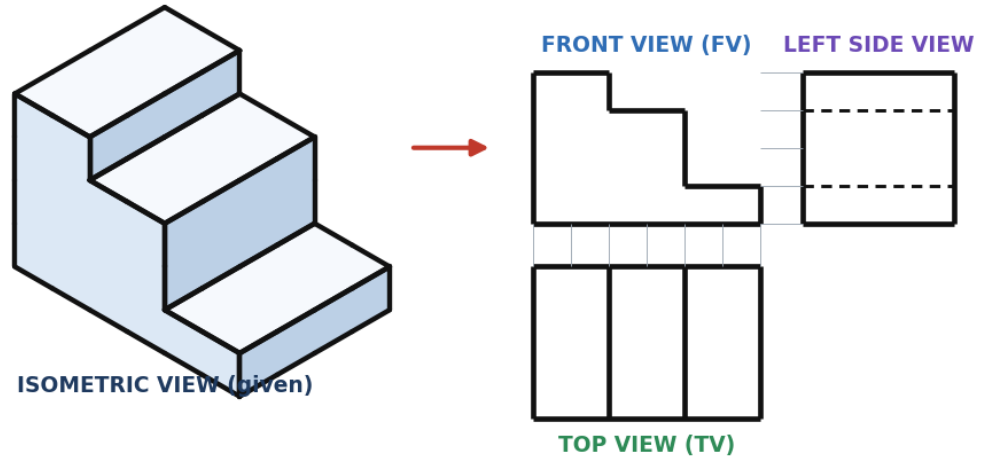
1. Study the isometric view: identify X, Y, Z directions
2. Choose FV direction (the most descriptive face)
3. Draw FV: heights from the vertical edges, widths ∥ X
4. Project TV below FV and LSV to the right of FV
5. Add hidden lines (dashed) & centre lines
6. Dimension & label

Orthographic → Isometric

1. Read the 3 views together: find overall L × B × H
2. Draw the ENCLOSING BOX in isometric (30° lines)
3. Mark features from FV on the front face, from TV on the top face
4. Cut / add portions; draw edges ∥ to iso axes only
5. Circles → ellipses; slant lines → join end points
6. Erase construction lines

Fig. 3.1 — Procedure for conversion in both directions

Example 1 – stepped block



Steps in FV; TV shows step lines; LSV: hidden (dashed) lines of lower steps

Fig. 3.2 — Example 1 – stepped block

PROCEDURE

1. **FV direction:** choose the face that shows the **most shape** (here the left face that shows the steps). Draw FV with correct heights and widths.
2. **TV:** project down from FV; width stays the same, depth = length of the y-axis lines. Each **step** appears as a **line** in TV.
3. **LSV:** project right; **lower steps lie behind** the tallest face → shown by **hidden (dashed) lines**.
4. Check: every point on the isometric must appear in all three views at the aligned positions.

Example 2 – block with rectangular through-slot

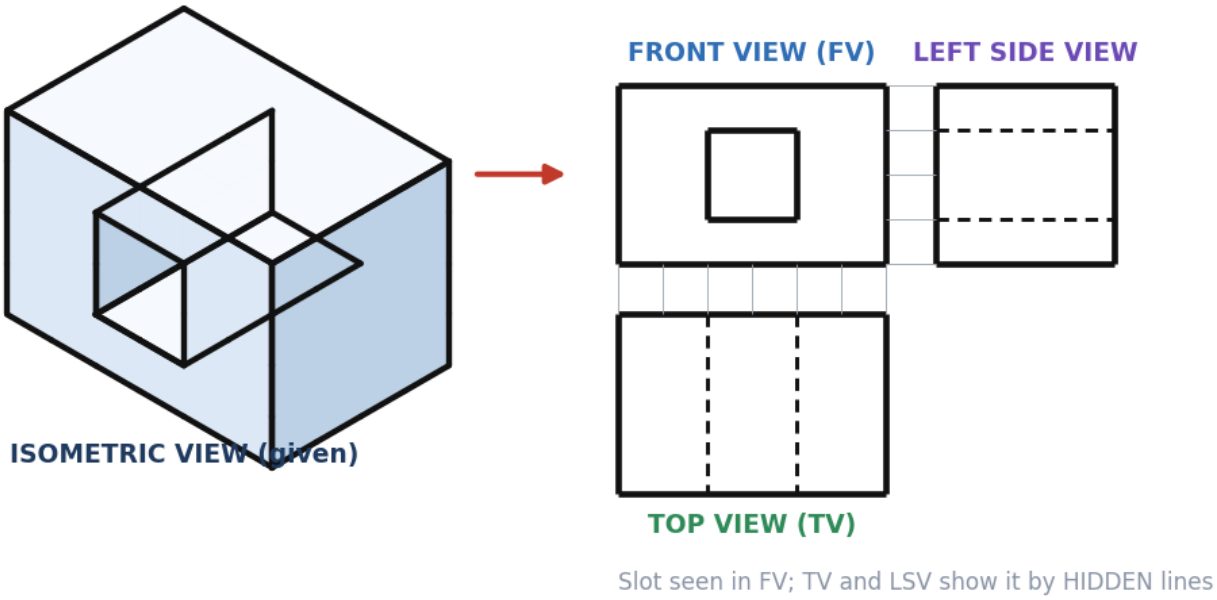


Fig. 3.3 — Example 2 – block with a rectangular through-slot

Example 3 – block with notch & step

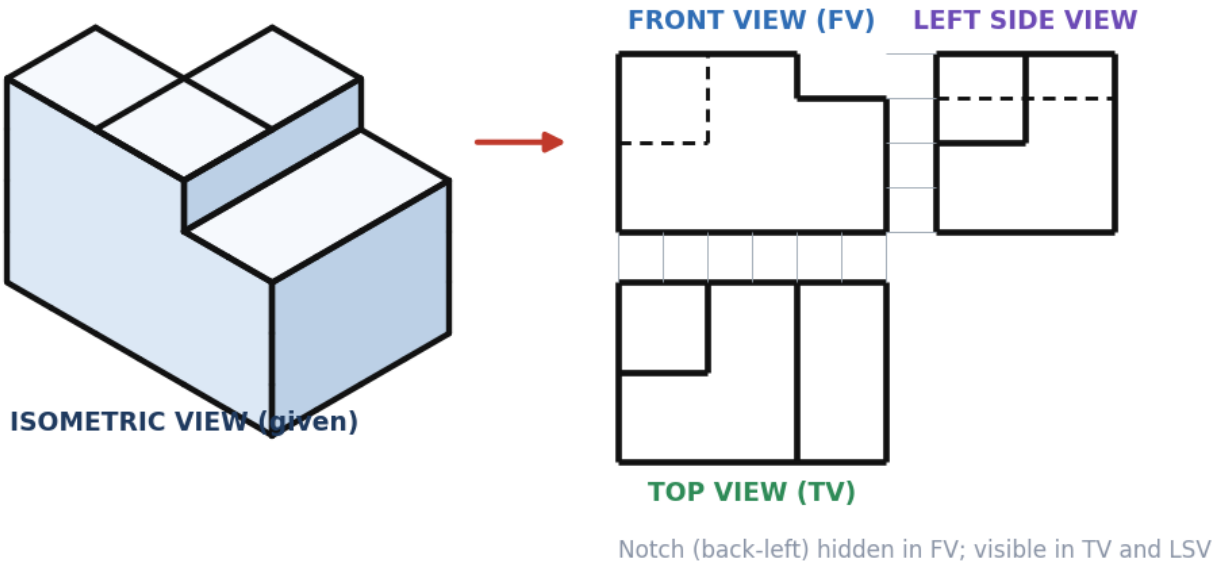


Fig. 3.4 — Example 3 – block with a notch (hidden in one view, visible in others)

Example 5 – inverted-T block with a top slot

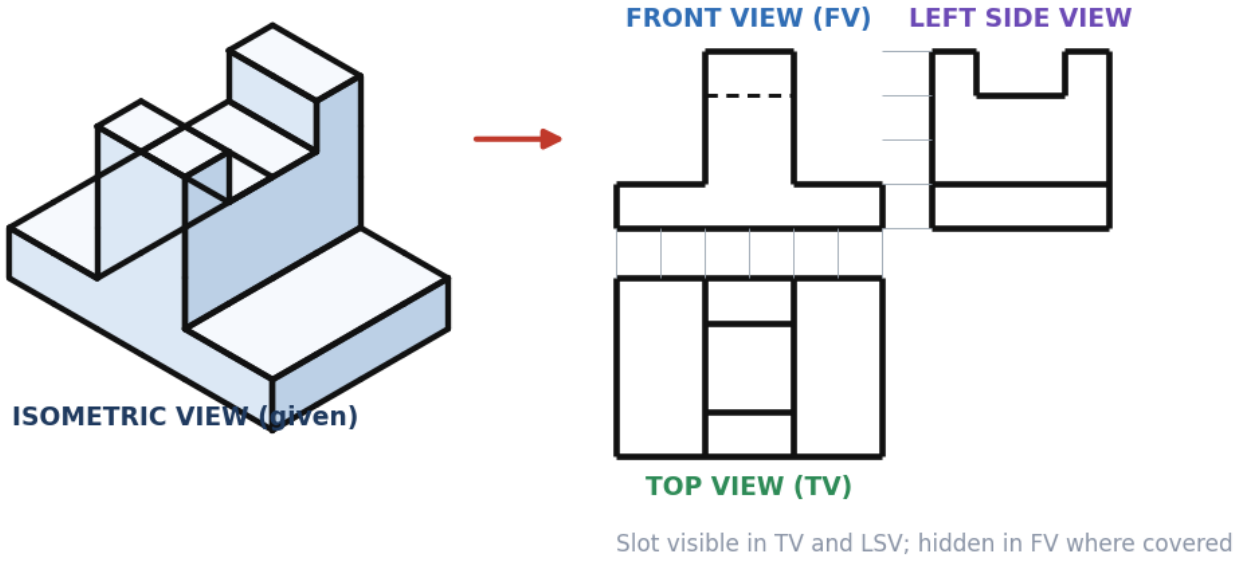


Fig. 3.5 — Example 4 – inverted-T block with a slot in the upright

KEY POINTS

- A feature that is **open to the observer** is drawn with **thick** lines; a feature **inside or behind** is drawn **dashed**.
- A through-slot appears **solid** in the view it passes through and **dashed** in the other two.
- Count the **faces**: the number of regions in each view should match the faces you can see from that direction.

4. Orthographic Views → Isometric View

Orthographic → Isometric (example)

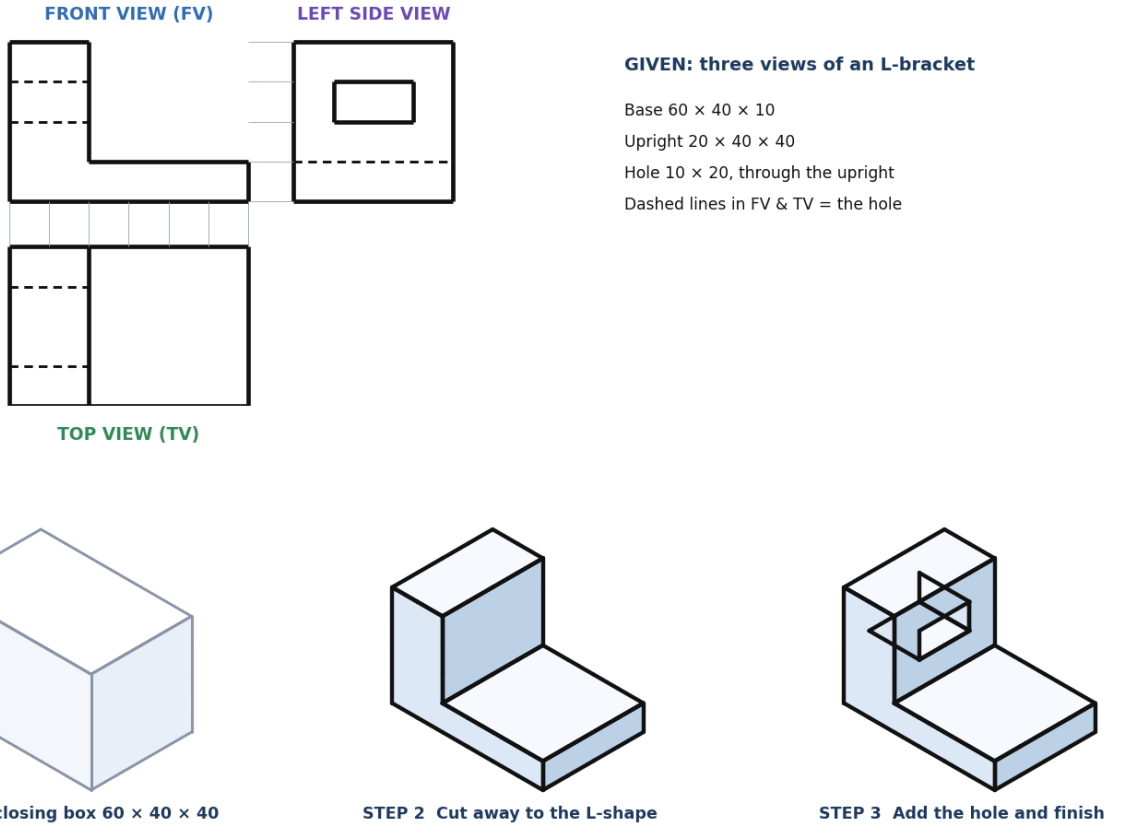


Fig. 4.1 — Example 5 – L-bracket with a through-hole

PROCEDURE

1. Find overall **L, B, H** from the views (FV gives L & H; TV gives L & B).
2. Draw the **enclosing box** using lines at **30°** (X and Y directions) and vertical (Z). Use thin construction lines.
3. Transfer features: **FV → front face** (x–z plane), **TV → top face** (x–y plane), **LSV → side face** (y–z plane). Measure along isometric lines only.
4. Remove material / add parts. For **inclined edges** locate their two end-points on isometric lines, then join them.
5. Make final edges thick; **do not show hidden lines** in an isometric view (unless required); erase construction lines.

Feature in orthographic views	In isometric
Step / shoulder	Offset along the iso lines by step height / depth
Through hole (rectangular)	Draw its opening on the face with iso lines; show the visible inner wall
Circular hole / boss	Isometric ellipse (four-centre) on that face
Inclined (chamfer / slope)	Non-isometric line: join two located end points
Symmetry line	Draw ‖ to iso axis through the centre

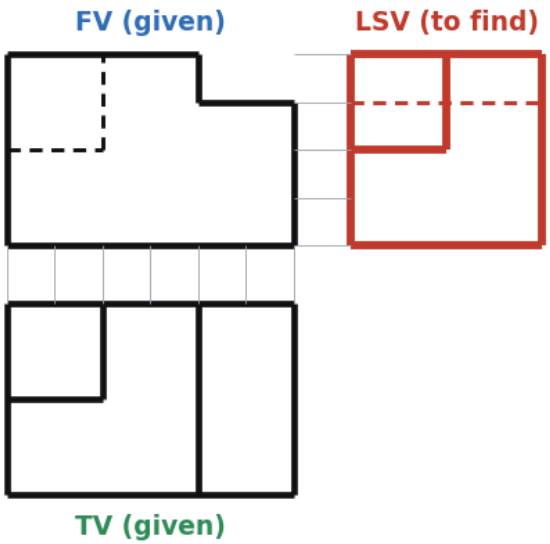
SOLVED EXAMPLE

Quick numerical check. Cube of side 40 mm: isometric *view* edge = 40 mm; isometric *projection* edge = 40 × 0.816 = **32.7 mm**; diagonal of a face (non-isometric line) is not measured – locate its end points.

APPLICATIONS
Conversions are used in **machine-part drawings, CAD model checking, assembly instructions, furniture and architectural plans** – to understand how a part looks and how it is made.

4.1 Missing-view problems

Missing-view problem: draw the LSV from the given FV and TV



Method: take HEIGHTS from the FV, take DEPTHS from the TV (measured from the XY side = back of object), and show the hidden notch with dashes.

Fig. 4.2 — Given two views, find the third

Visualisation tip	How it helps
Number the corners in all views	Tracks every point between FV, TV and LSV
Think of each closed region as a face	A region in one view = a surface; decide whether it is horizontal, vertical or inclined from the other view
A line in one view with a point in another	A face ⊥ to that plane / an edge ⊥ to the plane
Imagine cutting the object with the box lines	Helps to “see” the missing portion
Sketch a rough isometric first	Checks that all three views agree

5. Common Mistakes & Revision

- Using **angles** directly in isometric – always locate end points with lengths along the three axes.
- Placing the TV **above** the FV, or the side view on the wrong side (this is third-angle).
- Forgetting **hidden lines** or drawing them where the feature is visible.
- Confusing **isometric scale** (0.816) with the **isometric axes angle** (30° / 120°).
- Drawing circles as true circles in isometric – they must be **ellipses**.

QUICK SUMMARY

- Isometric: axes 120°, lines ∥ axes true, scale 0.816, circles → ellipses, no hidden lines.
- First angle: TV below FV, LSV on the right of FV; front of object = bottom of TV.
- Iso → ortho: choose FV, project TV & LSV, add dashed lines.
- Ortho → iso: overall box → features from each view → remove / add → thick final lines.

EXAM-ORIENTED QUESTIONS

Q1. Define isometric axes, isometric lines and isometric scale. State the value of the isometric scale. **[4]**

Q2. Explain the difference between isometric view and isometric projection. **[4]**

Q3. Draw the FV, TV and LSV of the object whose isometric view is shown (stepped block with a slot). **[8]**

Q4. Draw the isometric view of an object whose three orthographic views are given (L-bracket / T-block). **[8]**

Q5. Draw an isometric view of a cylinder Ø40 × 60 standing on its base; show the isometric circles on top. **[6]**

Q6. List the rules for drawing hidden lines in orthographic views. **[6]**

Q7. Draw the views of a block 60 × 40 × 40 with a 20 × 20 notch at one top corner. **[8]**